Canonical Derivative Currents and Covariance-Normal Energy for a Three-Component Fourth-Order Gaussian Field

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Abstract

We study three complex Gaussian fields on a fixed three-dimensional torus, with a positive matrix-valued fourth-order covariance and a fixed positive density floor. We construct a rational one-form derivative current by a common spectral approximation and construct its quadratic energy by exact conditional covariance subtraction. The current converges in every finite probability moment with spatial regularity below $-1/2$; the centered energy converges in $L^2(\Omega; H^{-1-\kappa})$ for each $\kappa > 0$. An explicit finite Gaussian integration-by-parts identity reduces the energy covariance to three connection families and treats the rational coefficient without a Taylor or Hermite truncation. For embedded Pauli generators we derive the exact leading contraction, sharp pointwise compensation slopes, a subcritical homogeneous instability, and a noncoercive critical boundary. A determinant estimate supplies a separate bound for deterministic positive backgrounds. The construction uses a fixed Gaussian reference; construction of a self-interacting Gibbs measure is outside the conclusions.

1 Introduction and position relative to previous work

Derivative composites at the regularity considered here require more than classical multiplication. Paracontrolled calculus supplies the commutator and paralinearization tools for the first composite [1]. Gaussian contraction and integration by parts supply the second. These mechanisms are established mathematics; our contribution is their explicit combination for the fixed rational current below, a regulator comparison for its conditionally centered energy, and the exact model-dependent stability boundaries. The Gaussian rough-path constructions of [2] are useful comparators, but a one-parameter rough-path theorem is not used to identify this three-dimensional distribution with a pathwise stochastic integral. We use the term canonical spectral current throughout.

The analytic BPHZ theorem [3] is a general stochastic convergence framework. Its hypotheses are not substituted for the direct estimates given here. In particular our coefficient is kept as an ordinary smooth function of the value field, rather than expanded into a globally convergent power series, and no interacting measure theorem is imported. The fixed reference, regulator class, and positive floor are part of every assertion. The companion dependency audit identifies standard inputs and the remaining model-specific propositions. Bounded literature searches do not certify priority or replace specialist novelty review.

Related Gaussian gradient-square and analytic Wick-functional results [4, 5] concern different covariance orders, rescalings or correlation limits. They do not directly supply the cross-cutoff Sobolev convergence for the ordinary rational matrix coefficient in this paper. The geometric stochastic heat equation [6] also treats nonlinear derivative composites, in a different dynamical and spatial setting. These comparisons rule out a claim to have introduced a general method for field-dependent derivative squares.

2 Fixed model and conventions

Let $L = 16$, $V = L^3$, $k_n = 2\pi n/L$, and write $\Psi(x) = \sum_{n\in\mathbb{Z}^3} \widehat{\Psi}(n) e^{ik_n\cdot x}$. The field is a centered stationary proper complex Gaussian with

$$\mathbb{E}[\widehat{\Psi}(n)\widehat{\Psi}(m)^\dagger] = \frac{2}{V}\mathcal{A}(k_n)^{-1}\mathbf{1}_{n=m}, \qquad \mathbb{E}[\widehat{\Psi}(n)\widehat{\Psi}(m)^T] = 0, \tag{1}$$

where

$$\begin{aligned} \mathcal{A}(k) &= (r + Z|k|^2 + Y|k|^4)I_3 + M, \\ M &= \mathrm{diag}(0, 0.03, 0.07) + 0.15(I_3 - \mathbf{1}\mathbf{1}^T/3), \\ (r, Z, Y) &= (0.4740336473, -0.9252754126, 1). \end{aligned} \tag{2}$$

These decimals are exact model inputs. Since $M \geq 0$ and $r - Z^2/(4Y) > 0$, the symbol is positive. It is comparable to $(1 + |k|^2)^2 I_3$. Realification means $X = (\mathrm{Re}\,\Psi, \mathrm{Im}\,\Psi) \in \mathbb{R}^6$; using $e_n(x) = V^{-1/2}e^{ik_n\cdot x}$, its orthonormal-mode covariance is realify$(\mathcal{A}^{-1})$. The factor two in (1) is therefore essential.

Let T_1, T_2, T_3 be the Pauli matrices embedded in the first two coordinates of $\mathbb{C}^3$ and let $S_A = \left(\begin{smallmatrix} \Re T_A & -\Im T_A \\ \Im T_A & \Re T_A \end{smallmatrix}\right)$. All S_A are real symmetric. Fix $\varepsilon = 10^{-12}$ and set

$$\rho = |X|^2, \quad m_A = X^T S_A X, \quad q_A = \frac{m_A}{\rho + \varepsilon}, \quad p_A = 2S_A X, \quad v_A = 2(S_A - q_A I)X. \tag{3}$$

The one-form $v_A \cdot dX = dm_A - q_A\, d\rho = (\rho + \varepsilon)dq_A$ is smooth with polynomially growing derivatives at this fixed floor. Define the positive coefficient matrix

$$B(X) = \sum_{A=1}^{3} \{a p_A p_A^T + b(p_A v_A^T + v_A p_A^T) + c v_A v_A^T\}. \tag{4}$$

Here $a = 0.2(0.3)^2/d$, $b = 0.1(0.3)(0.25)/d$, $c = 0.15(0.25)^2/d$, and $d = 2^2 + 10^{-12}$. Thus $a, c > 0$ and $ac - b^2 > 0$. Rounded values of a, b, c are $0.0045, 0.001875, 0.00234375$; calculations retain d exactly.

An admitted approximation has one real even scalar multiplier $R_N(n)$ on every real component, with $\sup_{N,n} |R_N(n)| < \infty$, $R_N(n) \to 1$ for each n, and finite support or sufficient rapid decay to make $X_N = R_N X$ smooth. All nonlinear products are actual products of these smooth fields, with exact Fourier convolution. This includes sharp cubes, sharp balls and scaled smooth cutoffs. No approximation rate is imposed on a general admitted sequence.

3 Canonical spectral derivative current

Fix an even smooth dyadic partition, use $f \prec g$ for its low–high paraproduct and $f \circ g$ for its comparable-frequency remainder. Write $\mathcal{C}^\alpha = B^\alpha_{\infty,\infty}$.

Lemma 3.1 (Dyadic reconstruction)**.** *Let $1/3 < \alpha < 1/2$, $x_N : \mathbb{T}^3_L \to \mathbb{R}^d$ with $d = 6$ here, and $f : \mathbb{R}^d \to (\mathbb{R}^d)^*$. Fiber indices below range from 1 to d. Suppose smooth approximations x_N converge in $\mathcal{C}^\alpha$ and their resonances $x_N^b \circ \partial_i x_N^a$ converge in $\mathcal{C}^{2\alpha-1}$. For every smooth one-form f with polynomially growing derivatives, $f_a(x_N)\partial_i x_N^a$ has a limit in $\mathcal{C}^{\alpha-1}$. The reconstruction is locally continuous in these enhanced inputs, with polynomial bounds on bounded x-ranges for the one-forms (3).*

Proof. The standard dyadic estimates are $\|u \prec v\|_\beta \lesssim \|u\|_\infty \|v\|_\beta$ and $\|u \circ v\|_{\beta+\gamma} \lesssim \|u\|_\beta \|v\|_\gamma$ when $\beta + \gamma > 0$. Also $\|u \prec v\|_{\beta+\gamma} \lesssim \|u\|_\beta \|v\|_\gamma$ for $\beta < 0$. Taylor expansion on each low-frequency block yields $f_a(x) = \partial_b f_a(x) \prec x^b + R_a(x)$, with $R_a(x) \in \mathcal{C}^{2\alpha}$. The problematic resonance becomes

$$f_a(x) \circ \partial_i x^a = \partial_b f_a(x)(x^b \circ \partial_i x^a) + \mathcal{C}(\partial_b f_a(x), x^b, \partial_i x^a) + R_a(x) \circ \partial_i x^a,$$

where $\mathcal{C}(u, v, w) = (u \prec v) \circ w - u(v \circ w)$. Subtracting u at the high-frequency spatial scale in each block gives a factor $2^{-j\alpha}$; the remaining two blocks cost $2^{-j\alpha} 2^{-j(\alpha-1)}$. Summation of comparable blocks and their lower output frequencies gives the commutator in $\mathcal{C}^{3\alpha-1}$ when $2\alpha - 1 < 0 < 3\alpha - 1$. This is the commutator estimate of [1, Lemma 2.4]; its block proof also gives continuity for differences. The Taylor remainder times $\partial_i x$ has the same positive regularity. Multiplication of $\partial_b f_a(x) \in \mathcal{C}^\alpha$ with the supplied resonance is allowed since $3\alpha - 1 > 0$. The other paraproducts have regularity at least $\alpha - 1$. Localization to a bounded range of x proves the claim for polynomially growing derivatives; the fixed rational denominator makes the localized constants polynomial in that range. □

Theorem 3.2 (Canonical current)**.** *For every finite $p \geq 1$ and $\kappa > 0$, $K_{A,i,N} = v_A(X_N) \cdot \partial_i X_N$ converges in $L^p(\Omega; \mathcal{C}^{-1/2-\kappa})$. All admitted approximations give the same limit. There is an almost surely convergent extracted subsequence. No local divergent current subtraction is required.*

Proof. Write C^{val} for the value-covariance kernel. The covariance bound $|\widehat{C}^{\mathrm{val}}(n)| \lesssim \langle n \rangle^{-4}$ gives $\mathbb{E}|\Delta_j X(x)|^2 \lesssim 2^{-j}$. Gaussian moments, Bernstein's inequality with a sufficiently large finite spatial integrability exponent, and summation give convergence of X_N in every finite probability moment in $\mathcal{C}^\alpha$, $\alpha < 1/2$. Low modes converge by the multiplier assumption and the high dyadic blocks are uniformly summable in each slightly weaker norm.

The resonance $\mathbb{X}^{ba}_{i,N} = X^b_N \circ \partial_i X^a_N$ has zero expectation by evenness. Wick pairing and the resonant frequency support give

$$\mathbb{E}|\widehat{\mathbb{X}}^{ba}_{i,N}(n)|^2 \lesssim \sum_{u+v=n,\ |u| \asymp |v|} \langle u \rangle^{-4} \langle v \rangle^{-2} \lesssim \langle n \rangle^{-3}. \tag{5}$$

For bounded $|u| \lesssim |n|$ there are $O(\langle n \rangle^3)$ comparable terms; the remaining sum is bounded by $\sum_{|u| \gtrsim |n|} \langle u \rangle^{-6}$. For each output mode, dominated convergence gives the Cauchy property for two arbitrary admitted multiplier sequences coupled to the same field. Second-chaos moment bounds and the same dyadic argument give convergence in $L^p(\Omega; \mathcal{C}^{-\eta})$ for every $\eta > 0$. Choose $\alpha \in (1/3, 1/2)$ and $\eta < 1 - 2\alpha$. Lemma 3.1, polynomial moment localization and a slightly larger probability exponent yield the result. Choose α sufficiently close to $1/2$ for the desired κ. A summable-probability subsequence follows from convergence in L^p. Evenness gives $\mathbb{E}[X^a_N \partial_i X^b_N] = 0$ at every cutoff. □

Remark 3.3 (Quantifiers and approximation rates)**.** *The multiplier assumptions alone give no universal rate in N. Multiplying a sharp cutoff by $1 - 1/\log(N+2)$ preserves them but creates a slow error in fixed low modes. Quantitative tail estimates for normalized standard cutoffs cannot be extended to this general class. Almost sure convergence above is for an extracted subsequence; a full prescribed dyadic sequence requires an additional summable error estimate.*

The one-form is not generally exact: $d(v_A \cdot dX) = (\rho + \varepsilon)^{-1} d\rho \wedge dm_A$. For $m = x^2 - y^2$ it equals $-8xy(x^2 + y^2 + \varepsilon)^{-1} dx \wedge dy$. Thus a deterministic chain rule alone does not construct this current. The declared common regulator class is sufficient for the result. Auxiliary correlated-field anomaly examples do not establish necessity of that class for the particular diagonal-leading covariance (1).

4 Covariance-normal energy

Put $Y_{N,i} = \partial_i X_N$ and $\Gamma_{N,i} = \mathbb{E}[Y_{N,i} Y_{N,i}^T]$. The exact density is

$$e_N = \tfrac{1}{2} \sum_i Y_{N,i}^T B(X_N) Y_{N,i}, \quad C_N(X) = \tfrac{1}{2} \sum_i \mathrm{Tr}[B(X)\Gamma_{N,i}], \quad E_N = e_N - C_N(X_N). \tag{6}$$

Independence of the jointly Gaussian vectors $X_N(x)$ and $Y_{N,i}(x)$ implies $C_N(X) = \mathbb{E}[e_N(x) \mid X_N(x) = X]$. In complex notation

$$C_N(\Psi) = \sum_{i,A} \{ a U_A^\dagger D_{N,i} U_A + 2b \Re(U_A^\dagger D_{N,i} V_A) + c V_A^\dagger D_{N,i} V_A \}, \tag{7}$$

where $U_A = T_A \Psi$, $V_A = (T_A - q_A I)\Psi$, $D_{N,i} = \mathbb{E}[\partial_i \Psi_N \partial_i \Psi_N^\dagger]$, and $\Gamma_{N,i} = \frac{1}{2}\,\mathrm{realify}(D_{N,i})$.

Lemma 4.1 (Finite Gaussian identity)**.** *Let $(X, Y, \widetilde{X}, \widetilde{Y})$ be jointly centered Gaussian, with $\mathbb{E}[XY^T] = \mathbb{E}[\widetilde{X}\widetilde{Y}^T] = 0$. Write $P_{a\beta} = \mathbb{E}[Y_a \widetilde{X}_\beta]$, $R_{c\alpha} = \mathbb{E}[\widetilde{Y}_c X_\alpha]$, $Q_{ac} = \mathbb{E}[Y_a \widetilde{Y}_c]$ and let $\Gamma, \widetilde{\Gamma}$ be the two Y covariances. For smooth polynomial-growth coefficients, the expectation of $A_{ab}(X)(Y_a Y_b - \Gamma_{ab}) D_{cd}(\widetilde{X})(\widetilde{Y}_c \widetilde{Y}_d - \widetilde{\Gamma}_{cd})$ equals*

$$\begin{aligned} &(Q_{ac}Q_{bd} + Q_{ad}Q_{bc})\mathbb{E}[A_{ab} D_{cd}] \\ &+ \{P_{a\beta}(Q_{bc}R_{d\alpha} + Q_{bd}R_{c\alpha}) + P_{b\beta}(Q_{ac}R_{d\alpha} + Q_{ad}R_{c\alpha})\}\mathbb{E}[\partial_\alpha A_{ab} \partial_\beta D_{cd}] \\ &+ P_{a\beta}P_{b\gamma}R_{c\alpha}R_{d\delta}\mathbb{E}[\partial_{\alpha\delta} A_{ab} \partial_{\beta\gamma} D_{cd}]. \end{aligned} \tag{8}$$

Repeated indices are summed; coefficients on the right are evaluated at X and $\widetilde{X}$, respectively.

Proof. Apply $\mathbb{E}[Z_j F(Z)] = \sum_l \mathrm{Cov}(Z_j, Z_l)\mathbb{E}[\partial_l F(Z)]$ to the four displayed derivative variables. Centering cancels same-vertex Y pairs and the mixed same-vertex covariances are zero. Each remaining leg pairs with an opposite Y or differentiates the opposite coefficient. There are two pairings with two Q bridges, four choices with one Q and one P and R, and one coefficient-derivative term with two P and R. These are exactly (8). Polynomial growth justifies Gaussian integration by parts by smooth truncation. The identity differentiates either coefficient at most twice. □

Theorem 4.2 (Energy convergence)**.** *For every $\kappa > 0$, E_N converges in $L^2(\Omega; H^{-1-\kappa}(\mathbb{T}_L^3))$, independently of the admitted multiplier sequence. The separately centered J^2, JK, K^2 converge jointly in that space. Also $\int E_N\,dx$ converges in $L^2(\Omega)$ with mean zero.*

Proof. Use common underlying Fourier modes for all approximations. Cross-cutoff value covariances have Fourier coefficients bounded by $C\langle n\rangle^{-4}$; one-derivative and two-derivative cross-covariances P, R, Q are bounded by $C\langle n\rangle^{-3}$ and $C\langle n\rangle^{-2}$. Value covariances converge uniformly in space since $\sum_n \langle n\rangle^{-4} < \infty$ in dimension three. Expectations of products of coefficient derivatives through order two therefore converge uniformly: localize the jointly Gaussian values to a fixed ball, use continuity in the finite covariance matrix there, and remove the localization by polynomial moment bounds uniform in the cutoffs.

Let $a_t(n) = \langle n\rangle^{-t}$ and let G_s be the periodic positive Bessel kernel with Fourier coefficients $\langle n\rangle^{-2s}$, $s > 1$. Its heat-kernel representation proves positivity. Splitting convolution sums into the smaller-than-half-output region and dyadic complementary annuli gives

$$a_2 * a_2 \lesssim a_1, \qquad a_3 * a_3 * a_3 * a_3 \lesssim_\eta a_{3-\eta} \quad (\eta > 0). \tag{9}$$

Indeed convolution with the dimension-critical a_3 adds at most a logarithm; three such logarithms are bounded by $C_\eta \langle n\rangle^\eta$. The first estimate follows directly from the two regions $|u| \le |n|/2$ and $|n-u| \le |n|/2$ and the remaining annuli.

Expand $\int G_s|Q_{NM}|^2$ in Fourier series and bound each summand absolutely by $C\langle n\rangle^{-2s}(a_2 * a_2)(n)$. This majorant is summable exactly for $s > 1$. Dominated convergence on the double sum proves strong convergence of Q_{NM} in $L^2(G_s\,dx)$ as both cutoffs tend to infinity, not just a uniform norm bound. The fourfold version and the second estimate of (9) prove strong convergence of P_{NM}, R_{NM} in $L^4(G_s\,dx)$. Consequently Q^2, QPR and P^2R^2 converge strongly in $L^1(G_s\,dx)$, the middle assertion following by Holder's inequality.

Apply Lemma 4.1 with A, D equal to the coefficient matrices in (4). The preceding strong convergence and uniform coefficient convergence make each cross-cutoff energy covariance converge against G_s. Stationarity and Fourier duality identify its integral, up to the fixed normalization V, with the squared H^{-s} probability norm. Polarization of the four covariances for $E_N - E_M$ proves the Cauchy property. The same argument works for each summand of B and for two different regulator sequences. Pairing with the constant function gives the integrated assertion. Conditional centering proves zero mean. □

The argument excludes H^{-1}; it does not prove that this endpoint fails for the full rational coefficient. Centering is exact for each regulator; subtracting only the leading divergent term is a different prescription.

5 Explicit contraction and stability boundaries

For sharp cubes $|n|_\infty \le N$, let

$$\delta = \frac{I_\square}{6\pi^2 Y L}, \qquad I_\square = 24\int_0^1\int_0^1 \frac{du\,dv}{1+u^2+v^2}. \tag{10}$$

Riemann summation of the homogeneous leading symbol gives $D_{N,i}/N \to \delta I_3$; lower-order symbol corrections are summable after differentiation. To justify the Riemann sum near zero, remove a fixed small cube there, apply ordinary summation elsewhere, and bound its contribution by $C\eta$ uniformly after rescaling. The face formula for $I_\square$ follows from $\operatorname{div}(u/|u|^2) = |u|^{-2}$ and the divergence theorem. Numerically $\delta \simeq 0.01619898645075695$.

Set $s = |\Psi_1|^2 + |\Psi_2|^2$ and $\rho = |\Psi|^2$. The Pauli identities $\sum_A m_A^2 = s^2$, $\sum_A \|T_A\Psi\|^2 = 3s$ imply

$$\frac{C_N(\Psi)}{N} \longrightarrow \delta W_\varepsilon(\Psi), \qquad W_\varepsilon = hs - \frac{6bs^2}{\rho+\varepsilon} - \frac{3cs^2(\rho+2\varepsilon)}{(\rho+\varepsilon)^2}, \quad h = 9(a+2b+c). \tag{11}$$

The s^2 terms follow by expanding $\sum_A \Re(U_A^\dagger V_A) = 3s - s^2/(\rho+\varepsilon)$ and $\sum_A \|V_A\|^2 = 3s - s^2(\rho+2\varepsilon)/(\rho+\varepsilon)^2$.

Proposition 5.1 (Sharp pointwise slopes)**.** *For the positive production coefficients, put $\ell = 9a + 12b + 6c$. For all fields,*

$$\ell s \le W_\varepsilon \le hs, \qquad \inf_{\Psi: s(\Psi)>0} W_\varepsilon/s = \ell, \qquad \sup_{\Psi: s(\Psi)>0} W_\varepsilon/s = h. \tag{12}$$

The upper bound $ms \ge W_\varepsilon$ holds for every field if and only if $m \ge h$. At $m = h$ the nonnegative residual has no positive lower bound proportional to s, even when s is fixed and nonzero.

Proof. The upper residual is the sum of the last two nonnegative terms in (11). For the lower bound write

$$W_\varepsilon - \ell s = 6bs\left(1 - \frac{s}{\rho+\varepsilon}\right) + 3cs\left(1 - \frac{s(\rho+2\varepsilon)}{(\rho+\varepsilon)^2}\right) \ge 0,$$

since $0 \le s \le \rho$ and $\rho(\rho+2\varepsilon) \le (\rho+\varepsilon)^2$. Taking $\rho = s \to \infty$ proves sharpness of ℓ. Taking fixed $s > 0$ and $\rho \to \infty$ proves sharpness of h and the failure of a positive uniform proportional bound for $hs - W_\varepsilon$. □

Proposition 5.2 (Subcritical homogeneous instability)**.** *Let $t \to \infty$, $m < h$ and $\gamma = 1.62$. A leading subtraction $-tW_\varepsilon$ and compensation tms, together with the local sextic $\gamma\rho^3/6$ and fixed lower-order production terms, admit homogeneous fields whose energy density is at most $-c_m t^{3/2} + O(t)$ for some $c_m > 0$.*

Proof. Choose a fixed $x \in (0,1)$ small enough that $A_x = (h-m)x - (6b+3c)x^2 > 0$ and take $s = x\rho$. At large ρ, $W_\varepsilon - ms = A_x\rho + O(1)$. All derivative currents vanish for this homogeneous field. The local energy is $\gamma\rho^3/6 - tA_x\rho + O(\rho^2 + t)$. Choose $\rho = (2tA_x/\gamma)^{1/2}$. The leading value is $-2^{3/2}(tA_x)^{3/2}/(3\sqrt{\gamma})$, and the error is $O(t)$. □

Thus the minimal leading compensation slope for a family energy $\frac{1}{2}\Delta\mu_N s$ is $\Delta\mu_N/N \to 2h\delta$ when only leading pointwise cancellation is considered. We make no claim that a remainder merely denoted $O(1)$ repairs the exact finite-cutoff counterterm uniformly in all field directions. The threshold is a local compensation statement, not a full-field coercivity theorem.

6 A deterministic-background exponential bound

This separate result gives a useful positive stability estimate. Let $b(x)$ be a deterministic bounded positive semidefinite 18×18 matrix on the bundle of all six-real spatial derivatives. Write $\|b\|_\infty$ for its essential supremum operator norm. Let C be the Gaussian covariance operator and $L_b = \nabla^* b \nabla$ in quadratic-form sense. Set $T_b = C^{1/2} L_b C^{1/2}$, $C_N^{\mathrm{op}} = R_N C R_N$ and $T_{b,N} = R_N T_b R_N$ for the same common multiplier. The centered quadratic form is $Q_{b,N} = \frac{1}{2}(\langle X_N, L_b X_N\rangle - \mathrm{Tr}(C_N^{\mathrm{op}} L_b))$.

Proposition 6.1 (Explicit frozen-background bound)**.** *Let $M_R = \sup|R_N|$ and $c_0 = (r+Y-\sqrt{(r-Y)^2+Z^2})/4 > 0$. For every $p \ge 0$ the forms converge in $L^2(\Omega)$ and*

$$\begin{aligned} 0 \le \log \mathbb{E} e^{-pQ_{b,N}} &\le \frac{p^2}{4}\|T_{b,N}\|_{\mathrm{HS}}^2 \\ &\le \frac{3p^2 M_R^4 \|b\|_\infty^2}{2c_0^2}\left(\frac{L}{2\pi}\right)^4\left(4\pi^2 + \frac{\pi^4}{45}\right). \end{aligned} \tag{13}$$

The partition factors converge to $\det_2(I + pT_b)^{-1/2}$.

Proof. The smallest eigenvalue of $\left(\begin{smallmatrix} r & Z/2 \\ Z/2 & Y \end{smallmatrix}\right)$, and $1 + t^2 \ge (1+t)^2/2$, give $\mathcal{A}(k) \ge c_0(1 + |k|^2)^2 I$. Quadratic-form order gives $0 \le T_{b,N} \le M_R^2\|b\|_\infty C^{1/2}(-\Delta)C^{1/2}$. Eigenvalue comparison bounds the squared Hilbert–Schmidt norm by $6M_R^4\|b\|_\infty^2 c_0^{-2}\sum_{n\neq 0}|k_n|^{-4}$. Counting the $24j^2+2$ integer points on the shell $|n|_\infty = j$ gives $\sum_{n\neq 0}|n|^{-4} \le 24\zeta(2) + 2\zeta(4) = 4\pi^2 + \pi^4/45$. The limiting operator is Hilbert–Schmidt, and strong convergence of bounded multipliers implies its cutoff conjugations converge in that norm. Gaussian diagonalization yields

$$\log \mathbb{E} e^{-pQ_{b,N}} = \tfrac{1}{2}\sum_j [p\lambda_{j,N} - \log(1 + p\lambda_{j,N})].$$

Use $0 \le x - \log(1+x) \le x^2/2$ for $x \ge 0$ and Hilbert–Schmidt convergence. For completeness, along a positive operator segment the trace derivative of $F_p(T) = \frac{1}{2}\mathrm{Tr}[pT - \log(I+pT)]$ is bounded by $\frac{p^2}{2}\|T\|_{\mathrm{HS}}$ in the Hilbert–Schmidt dual norm. Integration along the segment gives continuity in that norm and the claimed regularized determinant limit. □

For a deterministic bounded u, one can take $b = I_3 \otimes B(u)$, since $\|B(u)\| \leq (12a + 48|b| + 48c)|u|^2$. Replacing u by the random field in this proof would change the Gaussian diagonalization step and is not justified.

7 Scope, reproducibility and outstanding review

The statements are about composites under (1), at fixed L and fixed ε. The exact phase field $\Psi(x) = e^{ik\cdot x}u$, with $k \in (2\pi/L)\mathbb{Z}^3$, has $J_A = K_A = 0$ while $W_\varepsilon(u) > 0$ when $s > 0$. Consequently the conditional contraction is not a deterministic rate function for the bare field law. Likewise L^2 convergence of the integrated centered energy does not imply uniform integrability of its exponential. The self-coupled negative-exponential estimate, an interacting Gibbs limit, the A13 constructive program and a physical vacuum are not conclusions.

The verification package separates exact algebra and finite Gaussian moment checks from the analytic proofs above. It retains original source hashes, tests the real/complex covariance factor, tests all three integration-by-parts families with nonconstant coefficients, and rejects altered coefficients and stronger scope assertions. It also records the slow-cutoff negative control and the critical third-component escape. Numerical checks are reproducibility and falsification aids; they are not substitutes for the analytic limits.

This version is undergoing internal content review. Independent human mathematical review and specialist novelty review have not been performed. The external-review handoff records their disposition separately from file integrity and code execution. Author declarations, venue choice and actual transmission require the author's decisions.

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